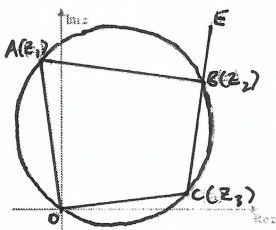


Complex Number 3

- Let a, b and c be real with $b^2 - 4ac < 0$, and suppose that the quadratic equation $az^2 + bz + c = 0$ has complex solutions $\alpha = x + iy$ and $\beta = u + iv$.
 - By considering the sum and product of the roots, show that $\text{Im}(\alpha + \beta) = 0$ and $\text{Im}(\alpha\beta) = 0$
 - Hence show that $\alpha = \bar{\beta}$.
- Graph the following loci:
 - $\text{Re}(z) = -3$
 - $\text{Im}(z) = 2$
 - $\text{Im}(z) < 1$
 - $\text{Im}(z) \geq -2$
 - $\text{Re}(z) = \text{Im}(z)$
 - $2\text{Re}(z) = \text{Im}(z)$
 - $\text{Re}(z) \leq 2\text{Im}(z)$
 - $\text{Re}(z) > -\text{Im}(z)$
- Let the point P represent the complex number $z = 2(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6})$, and let the points Q, R, S and T , represent $\bar{z}, -z, iz$ and $\frac{1}{z}$ respectively. Plot all these points on an Argand diagram.
- The number $z = a + ib$ and $w = iz$ are plotted in the complex plane at A and B respectively.
 - By considering the gradient, show that $OA \perp OB$,
 - Use the distance formula to show that $OA = OB$,
 - What type of triangle is $\triangle OAB$?
- The diagram shows three points A, B and C representing complex numbers z_1, z_2 and z_3 arranged on a circle through the origin O so that O, A, B and C are in clockwise order



You may assume that the arguments of all complex numbers z lie in the range $0 \leq \arg z < 2\pi$

- Explain why $\angle ABE = \arg(z_1 - z_2) - \arg(z_2 - z_3)$
 - Hence show that $z_1(z_2 - z_3) = kz_3(z_1 - z_2)$ where k is a positive real number
- Prove that $(1 + i \tan \theta)^n + (1 - i \tan \theta)^n = 2 \sec^n \theta \cos n\theta$
 - Hence prove that $\text{Re} \left(1 + i \tan \frac{\pi}{8} \right)^8 = 64(12\sqrt{2} - 17)$
 - Use de Moivre's theorem to express $\cos 4\theta$ and $\sin 4\theta$ in terms of $\cos \theta$ and $\sin \theta$.
 - Hence show that $\tan 4\theta = \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta}$.
 - Given $\alpha = \tan^{-1} \frac{1}{4}$, show that $4\alpha = \tan^{-1} \frac{240}{161}$.
 - Write $161 + 240i$ in the form $r(\cos \theta + i \sin \theta)$, expressing θ in terms of α .
 - Hence find, in the form $a + bi$ where a and b are integers, the four fourth roots of $161 + 240i$.
 - If $\left| \frac{z-1}{z+1} \right| = 2$, where $z = x + iy$, show that the equation of the locus of z is $\left(x + \frac{5}{3} \right)^2 + y^2 = \frac{16}{9}$.
 - Represent this locus on an Argand diagram and shade the region for which the inequalities $\left| \frac{z-1}{z+1} \right| \leq 2$ and $0 \leq \arg z \leq \frac{3\pi}{4}$ are both satisfied.

9. a. Show that $1 - \cos 2\theta - i \sin 2\theta = 2 \sin \theta (\sin \theta - i \cos \theta)$.

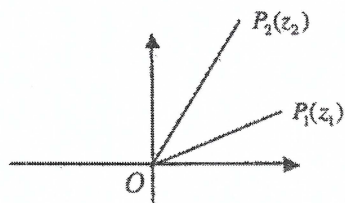
b. Given that $\frac{z-1}{z} = \operatorname{cis} \frac{2\pi}{5}$, show that $z = \frac{1}{2} \left(1 + i \cot \frac{\pi}{5} \right)$.

10. The complex numbers ω_1 and ω_2 have modulus 1, and arguments α_1 and α_2 respectively, where $0 < \alpha_1 < \alpha_2 < \frac{\pi}{2}$.

a. Draw a diagram showing all the given information.

b. Show that $\arg(\omega_1 - \omega_2) = \frac{1}{2}(\alpha_1 + \alpha_2 - \pi)$.

11. The points P_1 and P_2 represent the complex numbers z_1 and z_2 on the Argand diagram.



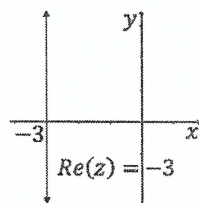
a. Prove that $|z_1 - z_2| \geq |z_1| - |z_2|$.

b. If $\left| z - \frac{4}{z} \right| = 2$ prove that the maximum value of $|z|$ is $\sqrt{5} + 1$.

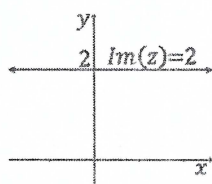
Answers:

2.

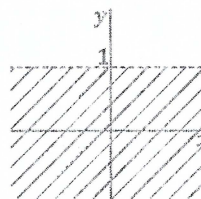
a.



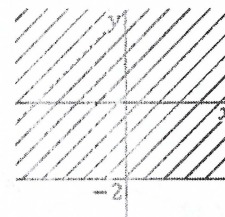
b.



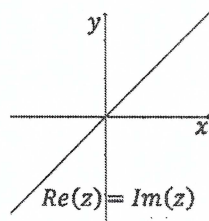
c.



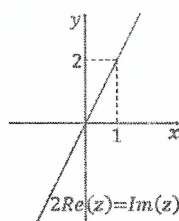
d.



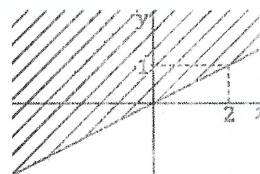
e.



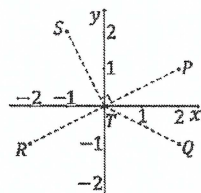
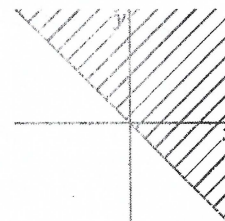
f.



g.



h.

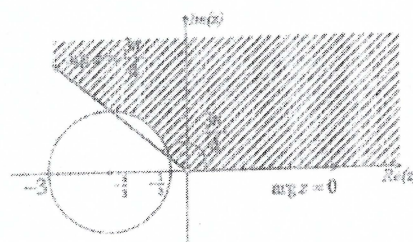


3.

4. c. right-isosceles

7. a. $\cos 4\theta = 8 \cos^4 \theta - 8 \cos^2 \theta + 1$, $\sin 4\theta = 4 \sin \theta \cos \theta - 8 \sin^3 \theta \cos \theta$ d. $289(\cos 4\alpha + i \sin 4\alpha)$ e. $\pm(4 + i), \pm(1 - 4i)$

8.



10.

